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Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

Wronskian of the function $y_c = c_1 + c_2 \cos x + c_3 \sin x$ is

Answer (Please select your correct option)

- 0
- 1

correct answer solve
by hadi

- 2
- 3

Made By: Waqar Siddhu

Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

The value of amplitude in the solution X=40Sin (7t+4) is

Answer (Please select your correct option)

- 7
- 4
- 40

correct answer solve
by hadi

- 20

Made By: Waqar Siddhu

Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

The degree of the differential equation $[1 - x(\frac{dy}{dx})^2]^{\frac{3}{2}} = \frac{d^2y}{dx^2}$ is

Answer (Please select your correct option)

- 1
- 2

correct answer solve
by hadi

- 3
- None of these

Made By: Waqar Siddhu

Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

Which of the following is the equation of pendulum?

Answer (Please select your correct option)

- $\frac{d^2\theta}{dt^2} + \frac{g}{l}\sin\theta = F(t)$

correct answer solve
by hadi

- $\frac{d\theta}{dt} + \frac{g}{l}\sin\theta = F(t)$
- $\frac{du}{dt} = F(u)G(t)$
- None of these

Made By: Waqar Siddhu

Question No : 5 of 26 Marks: 1 (Budgeted Time 1 Min)

The differential equation $(x^2 - 2x + 2y^2)\,dx + 2xy\,dy = 0$ is a/an -----differential equation.

Answer (Please select your correct option)

- Exact
- Non-exact

correct answer solve
by hadi

- Homogenous
- Separable

Made By: Waqar Siddhu

Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

In a Bernoulli equation $\frac{dy}{dx}+\frac{1}{x}y = xy^2$, *identify* $p(x), q(x)$ & n respectively.

Answer (Please select your correct option)

- $\frac{1}{x}, x$ & 2

correct answer solve
by hadi

- y, x^2y & 2
- $y, \frac{1}{x}$ & 2
- $\frac{1}{y}, y$ & 2

Made By: Waqar Siddhu

Question No : 7 of 26 Marks: 1 (Budgeted Time 1 Min)

If a differential equation is of the form $y\,f(x,y)dx + xg(x,y)\,dy = 0$ with $xM - yN \neq 0$, then the Integrating factor is------.

Answer (Please select your correct option)

- $\mu = \frac{1}{yN - xM}$
- $\mu = \frac{1}{xM + yN}$
- $\mu = \frac{1}{xM - yN}$
- $\mu = \frac{1}{yM + xN}$

correct answer solve
by hadi

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Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

If the given differential equation is homogeneous and $xM + yN \neq 0$, then the integrating factor of the given differential equation is given by------.

Answer (Please select your correct option)

- $u = \frac{1}{xM + yN}$

correct answer solve
by hadi

- $u = \frac{1}{xM - yN}$
- $u = \exp\left(\int \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{\quad} dy\right)$
- $u = \exp\left(\int \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{\quad} dx\right)$

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Question No : 9 of 26 Marks: 1 (Budgeted Time 1 Min)

Number of constant solutions obtained from the separable differential equation $\frac{dy}{dx} = \frac{y^3 - 1}{x}$ is---.

Answer (Please select your correct option)

- 0

correct answer solve
by hadi

- 1
- 2
- 3

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Question No : 10 of 26 Marks: 1 (Budgeted Time 1 Min)

If $\frac{dy}{dx} = x$, *then* $y =$ .....

Answer (Please select your correct option)

- $y = \frac{x^2}{2} + c$

correct answer solve
by hadi

- $y = x^2 + c$
- $y = x + c$
- $y = \frac{x}{2} + c$

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Question No : 11 of 26 Marks: 1 (Budgeted Time 1 Min)

To convert the non-homogenous differential equation $\frac{dy}{dx} = \frac{2x+3y-4}{4x+6y+9}$ into variable separable form, we use the substitution -----.

Answer (Please select your correct option)

- $Z = x + y$
- $Z = 4x + 3y$
- $Z = 2x + 6y$
- $Z = 2x + 3y$

correct answer solve
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Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

In exponential model for the population growth $P(t) = P_0 e^{kt}$ *If* $k < 0$, *then* $\lim_{t \to \infty} P(t) = - - -$.

Answer (Please select your correct option)

- ∞
- 0

correct answer solve
by hadi... k >0 then
infinte

- $-\infty$
- 1

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Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

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Answer (Please select your correct option)

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correct answer solve
by hadi

- $-\infty$
- 0
- 1

Made By: Waqar Siddhu

Question No : 14 of 26 Marks: 1 (Budgeted Time 1 Min)

The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are----.

Answer (Please select your correct option)

- P=0, P= b/a
- P=0, P=a/b

correct answer solve
by hadi

- P=a/b, P=b/a
- P=0,P=0

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Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

If $f(x)$ and $g(x)$ are linearly dependent on I, then-----.

Answer (Please select your correct option)

- $W(f,g)(x) \neq 0$ (Wronskian) for all x in the interval I.
- $W(f,g)(x) = 0$ (Wronskian) for all x in the interval I.

correct answer solve
by hadi

- $W(f,g)(x)$ *may or may not be* zero for all x in the interval I.
- $W(f,g)(x)$ is not defined for all x in the interval I.

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Question No : 16 of 26 Marks: 1 (Budgeted Time 1 Min)

What is annihilator operator of the function $g(x) = \sin x$?

Answer (Please select your correct option)

- $(D^3 + 1)$
- D^2
- $D^2 - 1$
- $D^2 + 1$

correct answer solve
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Question No : 17 of 26 Marks: 1 (Budgeted Time 1 Min)

Which of the following is the annihilator operator of the function $g(x) = 4x - 5$?

Answer (Please select your correct option)

- D^2
- $(D+4)$
- (D^3-4)
- $(D-4)$

correct answer solve
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Question No : 18 of 26 Marks: 1 (Budgeted Time 1 Min)

The differential equation of the orthogonal trajectory to the family of curves $y = cx$ is-----.

Answer (Please select your correct option)

- $\frac{dy}{dx} = -\frac{x}{y}$
- $\frac{dy}{dx} = \frac{x}{y}$
- $\frac{dy}{dx} = -\frac{y}{x}$
- $\frac{dy}{dx} = \frac{y}{x}$

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Question No : 19 of 26 Marks: 1 (Budgeted Time 1 Min)

If $y = 1 + 6x^2 - 7x^3$, then which of the following is true for it ?

Answer (Please select your correct option)

- Its annihilator operator is D.
- Its annihilator operator is D^2.
- Its annihilator operator is D^3.
- Its annihilator operator is D^4.

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Question No : 20 of 26 Marks: 1 (Budgeted Time 1 Min)

If $y_1 = x^2$ and $y_2 = x$ are the first and second solution of $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- $W(y_1, y_2)$ must be equal to zero on the indicated interval.
- $W(y_1, y_2)$ must be equal to non-zero on the indicated interval.

correct answer solve
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- $W(y_1, y_2)$ may or may not be equal to zero on the indicated interval.
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Answer (Please select your correct option)

- ∞
- 0
- $-\infty$
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- $(D-4)$

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- Its annihilator operator is D^2.
- Its annihilator operator is D^3.
- Its annihilator operator is D^4.

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Answer (Please select your correct option)

- $W(y_1, y_2)$ must be equal to zero on the indicated interval.
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Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

The solutions $y_1 = c_1$, $y_2 = c_2 \cos x$, $y_3 = c_3 \sin x$ of a Differential Equation are

Answer (Please select your correct option)

- Linearly independent
- Linearly dependent
- both of the other options

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Question No : 2 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Wronskian of the functions $y_1(x) = 1$, $y_2(x) = \text{Cos}x$, $y_3(x) = \text{Sin}x$ is

Answer (Please select your correct option)

- 0
- 1
- 2
- 3

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Question No : 3 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The value of amplitude in the solution X=7Sin (3t+ 2.45) is

Answer (Please select your correct option)

- 7
- 3
- 2.45
- 1

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Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

The order of the differential equation $(\frac{du}{dt})^{\frac{1}{3}} = \frac{1}{u}$ is

Answer (Please select your correct option)

1

2

3

4

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Question No : 5 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The solution of the differential equation $\frac{dy}{dx} = \tan x$ is ----------

Answer (Please select your correct option)

- $y = \ln|\sin x| + c$
- $y = \ln|\sec x| + c$
- $y = \ln|cosec\, x| + c$
- $y = \ln|\cos x| + c$

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Question No : 6 of 26 **Marks: 1 (Budgeted Time 1 Min)**

A differential equation $M(x, y)\,dx + N(x, y)\,dy = 0$ is said to be an exact if -----.

Answer (Please select your correct option)

- $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$
- $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$
- $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
- $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$

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Question No : 7 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The differential equation $(x^2 - 2x + 2y^2)\,dx + 2xy\,dy = 0$ is a/an -----differential equation.

Answer (Please select your correct option)

- Exact
- Non-exact
- Homogenous
- Separable

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Question No : 8 of 26 **Marks: 1 (Budgeted Time 1 Min)**

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t^2$; the integarating factor will be -----.

Answer (Please select your correct option)

- $\mu = -\frac{1}{t}$
- $\mu = \frac{1}{t}$
- $\mu = t$
- $\mu = 2t$

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Question No : 9 of 26 **Marks: 1 (Budgeted Time 1 Min)**

A differential equation $\frac{dy}{dx} + p(x)y = q(x)y^n \quad for\ n \neq 0,1$ is called a/an ----.

Answer (Please select your correct option)

- Non-Exact differential equation
- Linear differential equation
- Bernoulli equation
- Exact differential equation

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Question No : 10 of 26 **Marks: 1 (Budgeted Time 1 Min)**

In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^n$$

into linear differential equation, we choose ----.

Answer (Please select your correct option)

- $v = y^{n-1}$
- $v = y^{1-n}$
- $v = y^n$
- $v = y'$

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To convert the non-homogeneous differential equation $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$ into separable form, if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$, then we use the substitution-----.

Answer (Please select your correct option)

- $x = X + h$
- $y = Y + h$
- $z = a_1x + b_1y$
- $z = Y + X$

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Question No : 12 of 26 **Marks: 1 (Budgeted Time 1 Min)**

In exponential model for the population growth $P(t) = P_0 e^{kt}$ *If* $k > 0$, *then* $\lim_{t \to \infty} P(t) = - - -$.

Answer (Please select your correct option)

- ∞
- −∞
- 0
- 1

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Question No : 13 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are----.

Answer (Please select your correct option)

- P=0, P= b/a
- P=0, P=a/b
- P=a/b, P=b/a
- P=0,P=0

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Question No : 14 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The orthogonal trajectory to the family of curves $3x + 4y = c$ is------.

Answer (Please select your correct option)

- $y = 2x$
- $y - x = c$
- $4x - 3y = c$
- $4x + 3y = c$

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Question No : 15 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If the tangent lines of two curves are perpendicular at their point of intersection then both the curves are------

Answer (Please select your correct option)

- Non-intersecting curves
- Parallel curves
- Orthogonal curves
- Intersecting curves

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Question No : 16 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that K=0.043?

Answer (Please select your correct option)

- 371.415
- 380.560
- 363.279 — The formula is A(t)=Ao e^kt = 100(e^30*0.043)= **363.279**
- 360.351

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Question No : 17 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $y = 1 + 6x^2 - 7x^3$, then which of the following is true for it ?

Answer (Please select your correct option)

- Its annihilator operator is D.
- Its annihilator operator is D^2.
- Its annihilator operator is D^3.
- Its annihilator operator is D^4.

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Question No : 18 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $y = e^{-6x}$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- The annihilator operator of $y = e^{-6x}$ is $(D-6)^2$.
- The annihilator operator of $y = e^{-6x}$ is $(D-6)^3$.
- The annihilator operator of $y = e^{-6x}$ is $(D-6)$.
- The annihilator operator of $y = e^{-6x}$ is $(D+6)$.

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Question No : 19 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $y_1 = x$ and $y_2 = xe^x$ are the first and second solution of $x^2\frac{d^2y}{dx^2} - x(x+2)\frac{dy}{dx} + (x+2)y = 0$ on $(0,\infty)$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly dependent.
- $y_1 = x$ and $y_2 = xe^x$ must be linearly independent.
- $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly independent.
- $y_1 = x$ and $y_2 = xe^x$ must be linearly dependent.

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Question No : 20 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $y_2 = x^3$ is the second solution of $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 3y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- The first solution say y_1 must be non-trivial on the indicated interval.
- The first solution say y_1 can be trivial on the indicated interval.

correct answer solve
by hadi

- The first solution say y_1 must be discontinuous on the indicated interval.
- The first solution say y_1 may or may not be continuous on the indicated interval.

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Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

The form of the exact solution to

$$2\frac{dy}{dx} + 3y = e^{-x}, y(0) = 5$$

is

Answer (Please select your correct option)

- $Ae^{-1.5x} + Bxe^{-x}$
- $Ae^{1.5x} + Be^{-x}$
- $Ae^{1.5x} + Bxe^{-x}$
- $Ae^{-1.5x} + Be^{-x}$

correct answer solve
by hadi

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Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

The differential equation $2\frac{dy}{dx} + x^2 y = 2x + 3, y(0) = 5$ is

Answer (Please select your correct option)

- Linear

correct answer solve
by hadi

- Nonlinear
- Linear with fixed constants
- Undeterminable to be linear or nonlinear

Made By: Waqar Siddhu

Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

The value of amplitude in the solution X=40Sin (7t+4) is

Answer (Please select your correct option)

- 7
- 4
- 40
- 20

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

The order of the differential equation $(\frac{\partial^2 u}{\partial t^2})^3 + \frac{\partial u}{\partial t} = 0$ is

Answer (Please select your correct option)

- 0
- 1
- 2

correct answer solve
by hadi

- 3

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Question No : 5 of 26 Marks: 1 (Budgeted Time 1 Min)

Which of the following is the linear differential equation?

Answer (Please select your correct option)

- $x\frac{dy}{dx}+y=0$
- $y\frac{dy}{dx}+x=0$
- $x\frac{d^2y}{dx^2}+\frac{dy}{dx}+y^2=0$
- $\frac{dy}{dx}=e^y$

correct answer solve
by hadi

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Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

A solution obtained by giving particular values to the arbitrary constants in the General Solution of a differential equation is called a ----------

Answer (Please select your correct option)

- Singular solution
- Particular solution

correct answer solve
by hadi

- Explicit Solution
- None of these

Made By: Waqar Siddhu

Question No : 7 of 26 Marks: 1 (Budgeted Time 1 Min)

Which of the following equations satisfy the differential equation $\frac{dy}{dx} = x$

Answer (Please select your correct option)

- $2y = x^2 + c$

correct answer solve
by hadi

- $y = x^2 + c$
- $y = x^2$
- $y = x + c$

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Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

A differential equation $\frac{dy}{dx} + p(x)\,y = q(x)\,y^n \quad for\, n \neq 0,1$ is called a/an ----.

Answer (Please select your correct option)

- Non-Exact differential equation
- Linear differential equation
- Bernoulli equation

correct answer solve
by hadi

- Exact differential equation

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Question No : 9 of 26 Marks: 1 (Budgeted Time 1 Min)

In a Bernoulli equation $\frac{dy}{dx}+\frac{1}{x}y = xy^2$, *identify* $p(x), q(x)$ & n respectively.

Answer (Please select your correct option)

- $\frac{1}{x}, x$ & 2
- y, x^2y & 2
- $y, \frac{1}{x}$ & 2
- $\frac{1}{y}, y$ & 2

correct answer solve
by hadi

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Question No : 10 of 26 Marks: 1 (Budgeted Time 1 Min)

In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^n$$

into linear differential equation, we choose ----.

Answer (Please select your correct option)

- $v = y^{n-1}$
- $v = y^{1-n}$
- $v = y^n$
- $v = y'$

correct answer solve
by hadi

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Question No : 11 of 26 Marks: 1 (Budgeted Time 1 Min)

The differential equation ---------- is separable.

Answer (Please select your correct option)

- $x(x+y)\frac{dy}{dx}=4$
- $\frac{dy}{dx}=\frac{x^2}{x+xy}$
- $\frac{dy}{dx}=\frac{y}{1+xy^3}$
- $\frac{dy}{dx}=\frac{xy+3}{1+2xy}$

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

If $y = x$, then $\frac{dy}{dx} = ...$

Answer (Please select your correct option)

- 0
- 1

correct answer solve
by hadi

- x
- $x+1$

Made By: Waqar Siddhu

Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

If $f(x)$ and $g(x)$ are linearly dependent on I, then-----.

Answer (Please select your correct option)

- $W(f,g)(x) \neq 0$ (Wronskian) for all x in the interval I.
- $W(f,g)(x) = 0$ (Wronskian) for all x in the interval I.

correct answer solve
by hadi

- $W(f,g)(x)$ *may or may not be* zero for all x in the interval I.
- $W(f,g)(x)$ is not defined for all x in the interval I.

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Question No : 14 of 26 Marks: 1 (Budgeted Time 1 Min)

Wronskian of $Sin\ x, Cos\ x$ is----.

Answer (Please select your correct option)

- Sinx
- 1
- -1

correct answer solve
by hadi

- Cosx

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Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

The orthogonal trajectory to the family of curves $3x + 4y = c$ is------.

Answer (Please select your correct option)

- $y = 2x$
- $y - x = c$
- $4x - 3y = c$

correct answer solve
by hadi

- $4x + 3y = c$

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Question No : 16 of 26 Marks: 1 (Budgeted Time 1 Min)

If T denotes the half life and the initial amount of the substance be A_0 , then the amount of radioactive substance at time T is given by--------.

Answer (Please select your correct option)

- $A(T) = \frac{A_0}{2}$

correct answer solve
by hadi

- $A(T) = \frac{T}{2}$
- $A(T) = A_0$
- $A(T) = \frac{kT}{2}$

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Question No : 17 of 26 Marks: 1 (Budgeted Time 1 Min)

If the auxiliary equation has roots $-2, -2$, then the general solution of the differential equation is_____

Answer (Please select your correct option)

- $y = c_1 e^{-2x} + c_2 e^{-2x}$
- $y = c_1 e^{-2+2x}$
- $y = c_1 e^{2x} + c_2 e^{-2x}$
- $y = (c_1 + c_2 x) e^{-2x}$

correct answer solve
by hadi

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Question No : 18 of 26 Marks: 1 (Budgeted Time 1 Min)

If the auxiliary equation has complex roots $-2 \pm 3i$, then the general solution of the differential equation is_____

Answer (Please select your correct option)

- $y = ce^{2x}(\cos 10x + \sin 10x)$
- $y = e^{-2x}(c_1 \cos 3x + c_2 \sin 3x)$
- $y = e^{ix}(c_1 \cos 2x + c_2 \sin 2x)$
- $y = e^{2x}(c_1 \cos x - c_2 \sin x)$

correct answer solve
by hadi

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Question No : 19 of 26 Marks: 1 (Budgeted Time 1 Min)

If $y = e^{-6x}$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- The annihilator operator of $y = e^{-6x}$ is $(D-6)^2$.
- The annihilator operator of $y = e^{-6x}$ is $(D-6)^3$.
- The annihilator operator of $y = e^{-6x}$ is $(D-6)$.
- The annihilator operator of $y = e^{-6x}$ is $(D+6)$.

correct answer solve
by hadi

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Question No : 20 of 26 Marks: 1 (Budgeted Time 1 Min)

If the complementary solution of $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 3\sin x$ is $c_1 e^{-x} + c_2 e^{2x}$, then which of the following is the most accurate option for it?

Answer (Please select your correct option)

- The general form of the particular solution will be $(A\sin x + B\cos x)e^{x}$.
- The general form of the particular solution will be $Ax\sin x + Bx\cos x$.

correct answer solve
by hadi

- The general form of the particular solution will be $(Ax\sin x + Bx\cos x)e^{-x}$.
- The general form of the particular solution will be $A\sin x + B\cos x$.

Made By: Waqar Siddhu

Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

The form of the exact solution to

$$2\frac{dy}{dx} + 3y = e^{-x}, y(0) = 5$$

is

Answer (Please select your correct option)

- $Ae^{-1.5x} + Bxe^{-x}$
- $Ae^{1.5x} + Be^{-x}$
- $Ae^{1.5x} + Bxe^{-x}$
- $Ae^{-1.5x} + Be^{-x}$

correct answer solve
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Question No : 2 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The Period of oscillator in the solution X=50Sin (20t+8.5) is

Answer (Please select your correct option)

- 0.17643
- 0.32045
- 0.31400
- 0.58000

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Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

The Period of oscillator in the solution X=50Sin (20t+8.5) is

Answer (Please select your correct option)

- 0.17643
- 0.32045
- 0.31400
- 0.58000

Made By: Waqar Siddhu

Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

A problem of the differential equation associated with a single value of the unknown function at a given point in the domain of the solution is known as ----------

Answer (Please select your correct option)

- Boundary Value Problem
- Initial Value Problem
- None of these

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

The equation $\frac{dy}{dx} = \frac{x(x+1)}{y(y-1)}$ is a/an ----------

Answer (Please select your correct option)

- partial differential equation.

correct answer solve
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- ordinary differential equation.
- polynomial equation.
- transcendental equation.

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Question No : 5 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $f(x,y)=0$, then $\frac{dy}{dx}=....$

Answer (Please select your correct option)

- $-\frac{f_x}{f_y}$
- $-\frac{f_y}{f_x}$
- $\frac{f_x}{f_y}$
- $\frac{f_y}{f_x}$

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Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

For $y^2(1-x^2)-\cos^2 x = c$ *with* $y(0) = 3$, *the value of c is* ------.

Answer (Please select your correct option)

- 9
- 8
- 10
- 5

correct answer solve
by hadi

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Question No : 7 of 26 Marks: 1 (Budgeted Time 1 Min)

The differential equation $(3x^2y + 2)\,dx + (x^3 + y)\,dy = 0$ is -------.

Answer (Please select your correct option)

- Exact

  correct answer solve
  by hadi
- Non-exact
- Separable
- Homogenous

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Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t$; the continuous function, $q(t)$, is.....

Answer (Please select your correct option)

- $2t$
- $-\frac{1}{t}$
- $\frac{1}{t}$
- t

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Question No : 9 of 26 **Marks: 1 (Budgeted Time 1 Min)**

For a Ist order linear differential equation $\frac{dy}{dx}(x+2y^2)=y$; The continuous function, $q(y)$, is---.

Answer (Please select your correct option)

- x^2
- $2y$
- $2y^2$
- $-2y$

correct answer solve
by hadi

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Question No : 10 of 26 Marks: 1 (Budgeted Time 1 Min)

A differential equation $\frac{dy}{dx} + p(x)y = q(x)y^n \quad for\, n \neq 0,1$ is called a/an ----.

Answer (Please select your correct option)

- Non-Exact differential equation
- Linear differential equation
- Bernoulli equation

correct answer solve
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- Exact differential equation

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Question No : 11 of 26 Marks: 1 (Budgeted Time 1 Min)

The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are----.

Answer (Please select your correct option)

- P=0, P= b/a
- P=0, P=a/b

correct answer solve
by hadi

- P=a/b, P=b/a
- P=0,P=0

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Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

The logistic equation $\frac{dP}{dt} = P(a - bP)$ is a/an------.

Answer (Please select your correct option)

- Linear differential equation
- Non-linear differential equation

correct answer solve
by hadi

- Homogenous differential equation
- Exact differential equation

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Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

Wronskian of $Sin\ x, Cos\ x$ is----.

Answer (Please select your correct option)

- Sinx
- 1
- -1

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- Cosx

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Question No : 14 of 26 **Marks: 1 (Budgeted Time 1 Min)**

What is annihilator operator of the function $\frac{dy}{dx} + 4y = e^{2x}$?

Answer (Please select your correct option)

- $(D+4)(D-2)$
- $(D-2)$
- $(D+2)$
- $D\left(D^2-2\right)$

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Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

What is the annihilator operator of the function $g(x) = \cos 2x$?

Answer (Please select your correct option)

- $(D-4)$
- $(D+4)$
- (D^2+4)
- (D^3-4)

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Question No : 16 of 26 Marks: 1 (Budgeted Time 1 Min)

The differential equation of the orthogonal trajectory to the family of curves $y = cx$ is-----.

Answer (Please select your correct option)

- $\frac{dy}{dx} = -\frac{x}{y}$
- $\frac{dy}{dx} = \frac{x}{y}$
- $\frac{dy}{dx} = -\frac{y}{x}$
- $\frac{dy}{dx} = \frac{y}{x}$

correct answer solve
by hadi

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Question No : 17 of 26 Marks: 1 (Budgeted Time 1 Min)

The orthogonal trajectory to the family of curves $3x + 4y = c$ is------.

Answer (Please select your correct option)

- $y = 2x$
- $y - x = c$
- $4x - 3y = c$
- $4x + 3y = c$

correct answer solve
by hadi

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Question No : 18 of 26 Marks: 1 (Budgeted Time 1 Min)

In the population growth model, the solution of the differential equation is given by .....

Answer (Please select your correct option)

- $A(t) = ke^{kt}$
- $A(t) = A_0 e^{kt}$
- $A(t) = A_0 / 2$
- $A(t) = ke^{A_0 t}$

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Question No : 19 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $y_2 = x^3$ is the second solution of $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 3y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- The first solution say y_1 must be non-trivial on the indicated interval.
- The first solution say y_1 can be trivial on the indicated interval.

correct answer solve by hadi

- The first solution say y_1 must be discontinuous on the indicated interval.
- The first solution say y_1 may or may not be continuous on the indicated interval.

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Question No : 20 of 26 Marks: 1 (Budgeted Time 1 Min)

If the complementary solution of $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 3\sin x$ is $c_1e^{-x} + c_2e^{2x}$, then which of the following is the most accurate option for it?

Answer (Please select your correct option)

- The general form of the particular solution will be $(A\sin x + B\cos x)e^x$.
- The general form of the particular solution will be $Ax\sin x + Bx\cos x$.

correct answer solve
by hadi

- The general form of the particular solution will be $(Ax\sin x + Bx\cos x)e^{-x}$.
- The general form of the particular solution will be $A\sin x + B\cos x$.

Made By: Waqar Siddhu

Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

The solutions $y_1 = c_1$, $y_2 = c_2 \cos x$, $y_3 = c_3 \sin x$ of a Differential Equation are

Answer (Please select your correct option)

- Linearly independent

correct answer solve
by hadi

- Linearly dependent
- both of the other options

Made By: Waqar Siddhu

Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

Wronskian of the functions $y_1(x) = 1$, $y_2(x) = \text{Cos}x$, $y_3(x) = \text{Sin}x$ is

Answer (Please select your correct option)

- 0
- 1
- 2
- 3

Made By: Waqar Siddhu

Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

The value of amplitude in the solution X=7Sin (3t+ 2.45) is

Answer (Please select your correct option)

- 7

correct answer solve
by hadi

- 3
- 2.45
- 1

Made By: Waqar Siddhu

Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

The order of the differential equation $(\frac{du}{dt})^{\frac{1}{3}} = \frac{1}{u}$ is

Answer (Please select your correct option)

1

correct answer solve
by hadi

2

3

4

Made By: Waqar Siddhu

Question No : 5 of 26 Marks: 1 (Budgeted Time 1 Min)

The solution of the differential equation $\frac{dy}{dx} = \tan x$ is ----------

Answer (Please select your correct option)

- $y = \ln|\sin x| + c$
- $y = \ln|\sec x| + c$
- $y = \ln|\operatorname{cosec} x| + c$

correct answer solve
by hadi

- $y = \ln|\cos x| + c$

Made By: Waqar Siddhu

Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

A differential equation $M(x, y)\,dx + N(x, y)\,dy = 0$ is said to be an exact if -----.

Answer (Please select your correct option)

- $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$
- $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$
- $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
- $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 7 of 26 Marks: 1 (Budgeted Time 1 Min)

The differential equation $(x^2 - 2x + 2y^2)\,dx + 2xy\,dy = 0$ is a/an -----differential equation.

Answer (Please select your correct option)

- Exact
- Non-exact

correct answer solve
by hadi

- Homogenous
- Separable

Made By: Waqar Siddhu

Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t^2$; the integarating factor will be -----.

Answer (Please select your correct option)

- $\mu = -\frac{1}{t}$
- $\mu = \frac{1}{t}$

correct answer solve
by hadi

- $\mu = t$
- $\mu = 2t$

Made By: Waqar Siddhu

Question No : 9 of 26 Marks: 1 (Budgeted Time 1 Min)

A differential equation $\frac{dy}{dx} + p(x)y = q(x)y^n$ $for\ n \neq 0,1$ is called a/an ----.

Answer (Please select your correct option)

- Non-Exact differential equation
- Linear differential equation
- Bernoulli equation

correct answer solve
by hadi

- Exact differential equation

Made By: Waqar Siddhu

Question No : 10 of 26 Marks: 1 (Budgeted Time 1 Min)

In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^n$$

into linear differential equation, we choose ----.

Answer (Please select your correct option)

- $v = y^{n-1}$
- $v = y^{1-n}$
- $v = y^n$
- $v = y'$

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 11 of 26 Marks: 1 (Budgeted Time 1 Min)

To convert the non-homogeneous differential equation $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$ into separable form, if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$, then we use the substitution-----.

Answer (Please select your correct option)

- $x = X + h$
- $y = Y + h$
- $z = a_1x + b_1y$

correct answer solve
by hadi

- $z = Y + X$

Made By: Waqar Siddhu

Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

In exponential model for the population growth $P(t) = P_0 e^{kt}$ *If* $k > 0$, *then* $\lim_{t \to \infty} P(t) = - - -$.

Answer (Please select your correct option)

- ∞
- $-\infty$
- 0
- 1

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are----.

Answer (Please select your correct option)

- P=0, P= b/a
- P=0, P=a/b

correct answer solve
by hadi

- P=a/b, P=b/a
- P=0,P=0

Made By: Waqar Siddhu

Question No : 14 of 26 Marks: 1 (Budgeted Time 1 Min)

The orthogonal trajectory to the family of curves $3x + 4y = c$ is------.

Answer (Please select your correct option)

- $y = 2x$
- $y - x = c$
- $4x - 3y = c$
- $4x + 3y = c$

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

If the tangent lines of two curves are perpendicular at their point of intersection then both the curves are------

Answer (Please select your correct option)

- Non-intersecting curves
- Parallel curves
- Orthogonal curves
- Intersecting curves

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 16 of 26 Marks: 1 (Budgeted Time 1 Min)

If initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that K=0.043?

Answer (Please select your correct option)

- 371.415
- 380.560
- 363.279

correct answer solve
by hadi

- 360.351

Made By: Waqar Siddhu

Question No : 17 of 26 Marks: 1 (Budgeted Time 1 Min)

If $y = 1 + 6x^2 - 7x^3$, then which of the following is true for it ?

Answer (Please select your correct option)

- Its annihilator operator is D.
- Its annihilator operator is D^2.
- Its annihilator operator is D^3.
- Its annihilator operator is D^4.

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 18 of 26 Marks: 1 (Budgeted Time 1 Min)

If $y = e^{-6x}$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- The annihilator operator of $y = e^{-6x}$ is $(D-6)^2$.
- The annihilator operator of $y = e^{-6x}$ is $(D-6)^3$.
- The annihilator operator of $y = e^{-6x}$ is $(D-6)$.
- The annihilator operator of $y = e^{-6x}$ is $(D+6)$.

correct answer solve by hadi

Made By: Waqar Siddhu

Question No : 19 of 26 Marks: 1 (Budgeted Time 1 Min)

If $y_1 = x$ and $y_2 = xe^x$ are the first and second solution of $x^2\frac{d^2y}{dx^2} - x(x+2)\frac{dy}{dx} + (x+2)y = 0$ on $(0,\infty)$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly dependent.
- $y_1 = x$ and $y_2 = xe^x$ must be linearly independent.

correct answer solve
by hadi

- $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly independent.
- $y_1 = x$ and $y_2 = xe^x$ must be linearly dependent.

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Question No : 20 of 26 Marks: 1 (Budgeted Time 1 Min)

If $y_2 = x^3$ is the second solution of $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 3y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- The first solution say y_1 must be non-trivial on the indicated interval.
- The first solution say y_1 can be trivial on the indicated interval.
- The first solution say y_1 must be discontinuous on the indicated interval.
- The first solution say y_1 may or may not be continuous on the indicated interval.

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

A solution obtained by giving particular values to the arbitrary constants in the General Solution of a differential equation is called a ----------

Answer (Please select your correct option)

- Singular solution
- Particular solution

correct answer solve
by hadi

- Explicit Solution
- None of these

Made By: Waqar Siddhu

Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

The equation $\frac{dy}{dx} = \frac{x(x+1)}{y(y-1)}$ is a/an ----------

Answer (Please select your correct option)

- partial differential equation.

i think this ...because
in partial all value
write up and down ,,,,

- ordinary differential equation.
- polynomial equation.
- transcendental equation.

Made By: Waqar Siddhu

Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

A differential equation $M(x, y)\,dx + N(x, y)\,dy = 0$ is said to be non-exact if -----.

Answer (Please select your correct option)

- $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$
- $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$
- $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$
- $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Made By: Waqar Siddhu

Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t^2$; the integarating factor will be -----.

Answer (Please select your correct option)

- $\mu = -\frac{1}{t}$
- $\mu = \frac{1}{t}$

correct answer solve
by hadi

- $\mu = t$
- $\mu = 2t$

Made By: Waqar Siddhu

Question No : 5 of 26 Marks: 1 (Budgeted Time 1 Min)

In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^n$$

into linear differential equation, we choose ----.

Answer (Please select your correct option)

- $v = y^{n-1}$
- $v = y^{1-n}$
- $v = y^n$
- $v = y'$

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

The differential equation $\frac{dy}{dx} = \frac{x+2y-4}{2x+y-5}$ is not homogeneous. To convert it into homogeneous form, we use the substitution------.

Answer (Please select your correct option)

- $x = hX$
- $y = kY$
- $z = x + 2y$

correct answer solve
by hadi...but check it

- $x = X + h$, $y = Y + k$

Made By: Waqar Siddhu

Question No : 7 of 26 Marks: 1 (Budgeted Time 1 Min)

Number of constant solutions obtained from the separable differential equation $\frac{dy}{dx} = \frac{y^3 - 1}{x}$ is---.

Answer (Please select your correct option)

- 0

correct answer solve
by hadi

- 1
- 2
- 3

Made By: Waqar Siddhu

Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

In the 1st order linear differential equation, $\frac{dy}{dx} + \frac{y}{\ln x} = x^2$ find $p(x)$.

Answer (Please select your correct option)

- $\frac{y}{\ln x}$
- $\frac{1}{\ln x}$
- x^2
- $\frac{x^2}{\ln x}$

Made By: Waqar Siddhu

Question No : 9 of 26 Marks: 1 (Budgeted Time 1 Min)

For a Ist order linear differential equation $\frac{dy}{dx} + y = x$, the integrating factor $\mu(x)$ is---.

Answer (Please select your correct option)

- x
- 1

correct answer solve
by hadi

- e^x
- e

Made By: Waqar Siddhu

Question No : 10 of 26 Marks: 1 (Budgeted Time 1 Min)

In exponential model for the population growth $P(t) = P_0 e^{kt}$ *If* $k < 0$, *then* $\lim_{t \to \infty} P(t) = - - -$.

Answer (Please select your correct option)

- ∞
- 0
- $-\infty$
- 1

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 11 of 26 Marks: 1 (Budgeted Time 1 Min)

What is other name of Logistic Model?

Answer (Please select your correct option)

- Epidemic Model
- Euler Model
- Verhulst Pearl Model

correct answer solve
by hadi

- Gussain Model

Made By: Waqar Siddhu

Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

Wronskian of $Sin\ x, Cos\ x$ is----.

Answer (Please select your correct option)

- Sinx
- 1
- -1
- Cosx

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

What is annihilator operator of the function $\frac{dy}{dx} + 4y = e^{2x}$?

Answer (Please select your correct option)

- $(D+4)(D-2)$
- $(D-2)$
- $(D+2)$

correct answer solve
by hadi

- $D\left(D^2-2\right)$

Made By: Waqar Siddhu

Question No : 14 of 26 Marks: 1 (Budgeted Time 1 Min)

What is the annihilator operator of the function $g(x) = \cos 2x$?

Answer (Please select your correct option)

- $(D-4)$
- $(D+4)$
- (D^2+4) i think this
- (D^3-4)

Made By: Waqar Siddhu

Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

The differential equation of the orthogonal trajectory to the family of curves $y = cx$ is-----.

Answer (Please select your correct option)

- $\frac{dy}{dx} = -\frac{x}{y}$
- $\frac{dy}{dx} = \frac{x}{y}$
- $\frac{dy}{dx} = -\frac{y}{x}$
- $\frac{dy}{dx} = \frac{y}{x}$

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 16 of 26 Marks: 1 (Budgeted Time 1 Min)

If T denotes the half life and the initial amount of the substance be A_0 , then the amount of radioactive substance at time T is given by--------.

Answer (Please select your correct option)

- $A(T) = \frac{A_0}{2}$

correct answer solve
by hadi

- $A(T) = \frac{T}{2}$
- $A(T) = A_0$
- $A(T) = \frac{kT}{2}$

Made By: Waqar Siddhu

Question No : 17 of 26 Marks: 1 (Budgeted Time 1 Min)

If the auxiliary equation has complex roots $\pm i$, then the general solution of the differential equation is_____

Answer (Please select your correct option)

- $y = c_1 \cos x + c_2 \sin x$
- $y = ce^{2x}(\cos x + \sin x)$
- $y = e^{ix}(c_1 \cos x + c_2 \sin x)$
- $y = e^{x}(c_1 \cos x - c_2 \sin x)$

correct answer solve
by hadi...but check it

Made By: Waqar Siddhu

Question No : 18 of 26 Marks: 1 (Budgeted Time 1 Min)

If the complementary solution of $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 4e^{2x}$ is $c_1e^{2x} + c_2e^{3x}$, then which of the following is the most accurate option for it?

Answer (Please select your correct option)

- The general form of the particular solution will be Ae^{3x}.
- The general form of the particular solution will be Ae^{x}.
- The general form of the particular solution will be Ae^{2x}.
- The general form of the particular solution will be Axe^{2x}.

Made By: Waqar Siddhu

Question No : 19 of 26 Marks: 1 (Budgeted Time 1 Min)

If $c_1 e^x$ and $c_2 x e^x$ are the solutions of $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$, then which of the following is the most accurate option for $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x$?

Answer (Please select your correct option)

- Its general form of the particular solution will be Ae^x.
- Its general form of the particular solution will be Axe^x.
- Its general form of the particular solution will be Ax^2e^x.
- Its general form of the particular solution will be Ax^3e^x.

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Question No : 20 of 26 Marks: 1 (Budgeted Time 1 Min)

If $e^{2x}\left(c_1 \cos 3x + c_2 \sin 3x\right)$ is the solution of $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$, then which of the following is the most accurate option for $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = e^{2x}\sin 3x$?

Answer (Please select your correct option)

- Its general form of the particular solution will be $Ae^{2x} + B\sin x + C\cos x$.
- Its general form of the particular solution will be $e^{2x}\left(A\sin x + B\cos x\right)$.
- Its general form of the particular solution will be $e^{2x}\left(Ax\sin x + Bx\cos x\right)$.

correct answer solve
by hadi.h

- Its general form of the particular solution will be $e^{2x}\left(Ax\sin 3x + Bx\cos 3x\right)$.

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Question No : 1 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The form of the exact solution to

$$2\frac{dy}{dx} + 3y = e^{-x}, y(0) = 5$$

is

Answer (Please select your correct option)

- $Ae^{-1.5x} + Bxe^{-x}$
- $Ae^{1.5x} + Be^{-x}$
- $Ae^{1.5x} + Bxe^{-x}$
- $Ae^{-1.5x} + Be^{-x}$

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Question No : 2 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The Period of oscillator in the solution X=50Sin (20t+8.5) is

Answer (Please select your correct option)

- 0.17643
- 0.32045
- 0.31400
- 0.58000

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Question No : 4 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The equation $\frac{dy}{dx} = \frac{x(x+1)}{y(y-1)}$ is a/an ----------

Answer (Please select your correct option)

- partial differential equation.
- ordinary differential equation.
- polynomial equation.
- transcendental equation.

i guess

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Question No : 5 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $f(x,y)=0$, then $\frac{dy}{dx}=....$

Answer (Please select your correct option)

- $-\frac{f_x}{f_y}$
- $-\frac{f_y}{f_x}$
- $\frac{f_x}{f_y}$
- $\frac{f_y}{f_x}$

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Question No : 6 of 26 **Marks: 1 (Budgeted Time 1 Min)**

For $y^2(1-x^2)-\cos^2 x = c$ *with* $y(0) = 3$, *the value of c is* ------.

Answer (Please select your correct option)

- 9
- 8
- 10
- 5

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Question No : 7 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The differential equation $(3x^2y + 2)\,dx + (x^3 + y)\,dy = 0$ is -------.

Answer (Please select your correct option)

- Exact
- Non-exact
- Separable
- Homogenous

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Question No : 8 of 26 **Marks: 1 (Budgeted Time 1 Min)**

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t$; the continuous function, $q(t)$, is.....

Answer (Please select your correct option)

- $2t$
- $-\frac{1}{t}$
- $\frac{1}{t}$
- t

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Question No : 9 of 26 **Marks: 1 (Budgeted Time 1 Min)**

For a Ist order linear differential equation $\frac{dy}{dx}(x+2y^2)=y$; The continuous function, $q(y)$, is---.

Answer (Please select your correct option)

- x^2
- $2y$
- $2y^2$
- $-2y$

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Question No : 10 of 26 **Marks: 1 (Budgeted Time 1 Min)**

A differential equation $\frac{dy}{dx} + p(x)\,y = q(x)\,y^n \quad for\, n \neq 0,1$ is called a/an ----.

Answer (Please select your correct option)

- Non-Exact differential equation
- Linear differential equation
- Bernoulli equation
- Exact differential equation

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Question No : 11 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are----.

Answer (Please select your correct option)

- P=0, P= b/a
- P=0, P=a/b
- P=a/b, P=b/a
- P=0,P=0

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Question No : 13 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Wronskian of $Sin\ x, Cos\ x$ is----.

Answer (Please select your correct option)

- Sinx
- 1
- -1
- Cosx

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Question No : 14 of 26 **Marks: 1 (Budgeted Time 1 Min)**

What is annihilator operator of the function $\frac{dy}{dx} + 4y = e^{2x}$?

Answer (Please select your correct option)

- $(D+4)(D-2)$
- $(D-2)$
- $(D+2)$
- $D\left(D^2-2\right)$

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Question No : 15 of 26 **Marks: 1 (Budgeted Time 1 Min)**

What is the annihilator operator of the function $g(x) = \cos 2x$?

Answer (Please select your correct option)

- $(D-4)$
- $(D+4)$
- (D^2+4)
- (D^3-4)

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Question No : 16 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The differential equation of the orthogonal trajectory to the family of curves $y = cx$ is-----.

Answer (Please select your correct option)

- $\frac{dy}{dx} = -\frac{x}{y}$
- $\frac{dy}{dx} = \frac{x}{y}$
- $\frac{dy}{dx} = -\frac{y}{x}$
- $\frac{dy}{dx} = \frac{y}{x}$

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Question No : 17 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The orthogonal trajectory to the family of curves $3x + 4y = c$ is------.

Answer (Please select your correct option)

- $y = 2x$
- $y - x = c$
- $4x - 3y = c$
- $4x + 3y = c$

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Question No : 18 of 26 **Marks: 1 (Budgeted Time 1 Min)**

In the population growth model, the solution of the differential equation is given by .....

Answer (Please select your correct option)

- $A(t) = ke^{kt}$
- $A(t) = A_0 e^{kt}$
- $A(t) = A_0 / 2$
- $A(t) = ke^{A_0 t}$

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Question No : 19 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $y_2 = x^3$ is the second solution of $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 3y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- The first solution say y_1 must be non-trivial on the indicated interval.
- The first solution say y_1 can be trivial on the indicated interval.
- The first solution say y_1 must be discontinuous on the indicated interval.
- The first solution say y_1 may or may not be continuous on the indicated interval.

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Question No : 20 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If the complementary solution of $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 3\sin x$ is $c_1 e^{-x} + c_2 e^{2x}$, then which of the following is the most accurate option for it?

Answer (Please select your correct option)

- The general form of the particular solution will be $(A\sin x + B\cos x)e^{x}$.
- The general form of the particular solution will be $Ax\sin x + Bx\cos x$.
- The general form of the particular solution will be $(Ax\sin x + Bx\cos x)e^{-x}$.
- The general form of the particular solution will be $A\sin x + B\cos x$.

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Question: 1 (Marks: 1) Attempted Questions: 25 Total Questions: 26

In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^n$$

into linear differential equation, we choose ----.

Choices:

- $v = y^{n-1}$
- $v = y^{1-n}$
- $v = y^n$
- $v = y'$

If $y = x$, then $\frac{dy}{dx} = ...$

Choices:

- 0
- 1
- x
- $x+1$

Question: 3 (Marks: 1) Attempted Questions: 25 Total Questions: 26

The orthogonal trajectory to the family of curves $3x+4y=c$ is------.

Choices:

- $y=2x$
- $y-x=c$
- $4x-3y=c$
- $4x+3y=c$

If the auxiliary equation of a differential equation has two distinct real roots m_1 and m_2, then the general solution of the differential equation is_____

Choices:

- $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$
- $y = c_1 e^{m_1 x} + c_2 e^{-m_2 x}$
- $y = c_1 e^{-m_1 x} + c_2 e^{m_2 x}$
- $y = c_1 e^{m_1 + m_2 x}$

Question: 5 (Marks: 1) Attempted Questions: 25 Total Questions: 26

If the annihilator operator of $2x$ is D^2 and of xe^{3x} is $(D-3)^2$, then which of the following is the correct option?

Choices:

- The annihilator operator of $2x + xe^{3x}$ is $(D-3)^2$.
- The annihilator operator of $2x + xe^{3x}$ is D^2.
- The annihilator operator of $2x + xe^{3x}$ is $D^2 + (D-3)^2$.
- The annihilator operator of $2x + xe^{3x}$ is $D^2(D-3)^2$.

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If $\frac{dy}{dx} = \frac{1}{x}$, then which of the following is the most appropriate option for it?

Choices:

- $y = \ln x + c$
- $y = x^2$
- $y = \ln \sqrt{x} + c$
- $y = \ln \frac{1}{\sqrt{x}} + c$

$$\frac{dy}{dx} = \frac{2x+3y-4}{4x+6y+9}$$

If , then which of the following substitution is appropriate to make it a homogeneous differential equation?

Choices:

- $z = 2x + 3y$
- $z = x + y$
- $z = 4x + 3y$
- $z = 2x + 6y$

For a homogeneous differential equation of the form $y f(xy)dx + xg(xy)\,dy = 0$ and if $xM - yN \neq 0$ then the Integrating factor is.....

Choices:

- $\mu = \frac{1}{xM + yN}$
- $\mu = \frac{1}{xM - yN}$
- $\mu = \frac{1}{yM + xN}$
- None of these

How many initial conditions are necessary to determine the particular solution of a **first order** linear differential equation?

Choices:

- 0
- 1
- 2
- 3

Question: 10 (Marks: 1) Attempted Questions: 25 Total Questions: 26

If $y_1 = xe^{-x}$ is the first solution of $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$, then which of the following is true for $y_2 = y_1 \int \frac{e^{-2x}}{y_1^2}$ its second solution

Choices:

- $xe^{-x} \int \frac{2}{e^{-2x}} dx$
- $xe^{-x} \int \frac{2}{x^2 e^{-2x}} dx$
- $xe^{-x} \int \frac{1}{x^2 e^{-2x}} dx$
- $xe^{-x} \int \frac{1}{x^2} dx$

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Question: 11 (Marks: 1) Attempted Questions: 25 Total Questions: 26

Wronskian $W(x, x^2)$ is----

Choices:

- x
- x^2
- x^3
- 0

$$\frac{dy}{dx} + p(x)y = q(x)$$

In first order linear differential equation , the functions p(x) and q(x) must be .....

Choices:

- Continuous
- Discontinuous
- Homogeneous
- Non homogeneous

Question: 13 (Marks: 1) Attempted Questions: 25 Total Questions: 26

Differential of $(x+y)$ is -----.

Choices:

- $dx+dy$
- $dx-dy$
- $d(xy)$
- $d\left(\frac{x}{y}\right)$

Question: 14 (Marks: 1) Attempted Questions: 25 Total Questions: 26

Which of following is an example of boundary value problem (BVP) ?

Choices:

- $\frac{d^2y}{dx^2}+\lambda^2 x=0, \quad x(0)=x_0, x'(0)=0$
- $\frac{d^2y}{dx^2}+\lambda^2 x=0, \quad x(0)=x_0, x'(1)=0$
- $\frac{d^2y}{dx^2}+\lambda^2 x=0, \quad x(0)=x_0$
- $\frac{d^2y}{dx^2}+\lambda^2 x=0, \quad x'(1)=0$

For a 2nd order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 3, W_1 = -\frac{1}{4}$, then $u_1' = \cdots$.

Choices:

- 12
- -12
- $\frac{1}{12}$
- $-\frac{1}{12}$

Question: 16 (Marks: 1) Attempted Questions: 25 Total Questions: 26

If $y = a\sin x + b\cos x$ is the complementary solution of $y'' + y = 1$, then the general form of its particular solution is $y_p = \cdots$

Choices:

- A
- Ax
- $Ax + B$
- $Ax\sin x + Bx\cos x$

Question: 17 (Marks: 1) Attempted Questions: 25 Total Questions: 26

For some given functions, the Wronskian $W \neq 0$ somewhere in given interval, then given functions are linearly --------.

Choices:

- (●) Independent
- ( ) dependent

Question: 18 (Marks: 1) Attempted Questions: 25 Total Questions: 26

The value of amplitude in the solution of simple harmonic motion

$x(t) = 40\sin(7t + 1.86)$

is.....

Choices:

- 7
- 15
- 40
- 20

The orthogonal trajectory to the family of curves $x+2y=2$ is-----.

Choices:

- $y=-2x$
- $y-2x=c$
- $2x-3y=c$
- $4x+3y=c$

Differential Equations (MTH401)

Question: 20 (Marks: 1) **Attempted Questions: 25** **Total Questions: 26**

If the tangent lines of two curves are perpendicular at their point of intersection then both are-----

Choices:

- (●) Orthogonal
- ( ) Non-Orthogonal